

INTRODUCTION

- Persistent developmental stuttering (PDS)** associated with disrupted basal ganglia- and cerebellar- thalamocortical networks involved in speech sequencing and timing → their causal role remains unclear.
- Ventral intermediate (VIM) thalamic nucleus** deep brain stimulation has been associated with reduced stuttering in literature.
- Low-intensity focused ultrasound (LIFU)** enables noninvasive, spatially precise modulation of deep brain targets.
- We present an ongoing investigation of **VIM-LIFU in adults with PDS**.

METHODS

- Targeting:** MRI-guided neuronavigation to localize left VIM.
- Intervention:** Participants undergo active (VIM-LIFU) and sham (reversed transducer direction: no-stim) sessions with pre/post multimodal assessments (one week apart; pseudo-randomized).
- Outcomes:** fMRI assesses thalamocortical network changes; behavioral and speech measures assess motor timing, response inhibition, and speech fluency.

Study Question & Hypotheses

Research Question: Can noninvasive VIM LIFU neuromodulation alter the neural circuitry and motor-control processes underlying speech fluency in adults who stutter?

Multimodal Assessment

NEURAL: Does VIM-LIFU alter functional organization of the speech-motor network? *We assess both resting-state and task-evoked circuit hypotheses:*

H1: Resting-State Functional Connectivity → Active VIM-LIFU modulation will enhance VIM connectivity with cortical and subcortical CBGCTC speech-motor regions compared to sham.
H2: Task-Based Functional Connectivity (SST) → Active VIM-LIFU modulation will enhance VIM-IFG and VIM-STN connectivity during Successful Stop > Go compared to sham.

BEHAVIORAL: Does VIM-LIFU alter motor inhibition and motor timing that are relevant to fluent speech? *We assess two complementary processes:*

H3: Motor Inhibition - SST → Active VIM-LIFU will enhance motor inhibition, generating a greater decrease in SSRT from pre to post compared to sham.
H4: Motor Timing - BBA → Active VIM-LIFU will improve internal motor timing, generating a greater increase in rhythm-discrimination sensitivity (d') from pre to post compared to sham.

SPEECH: Does VIM-LIFU improve speech fluency? **H5: Speech Fluency** → Active VIM-LIFU will produce a greater reduction in stuttering severity (%SS) from pre- to post-stimulation than sham.

Experimental Setup & Preliminary Results

Behavioral: Stop Signal Task (SST: Motor Inhibition)

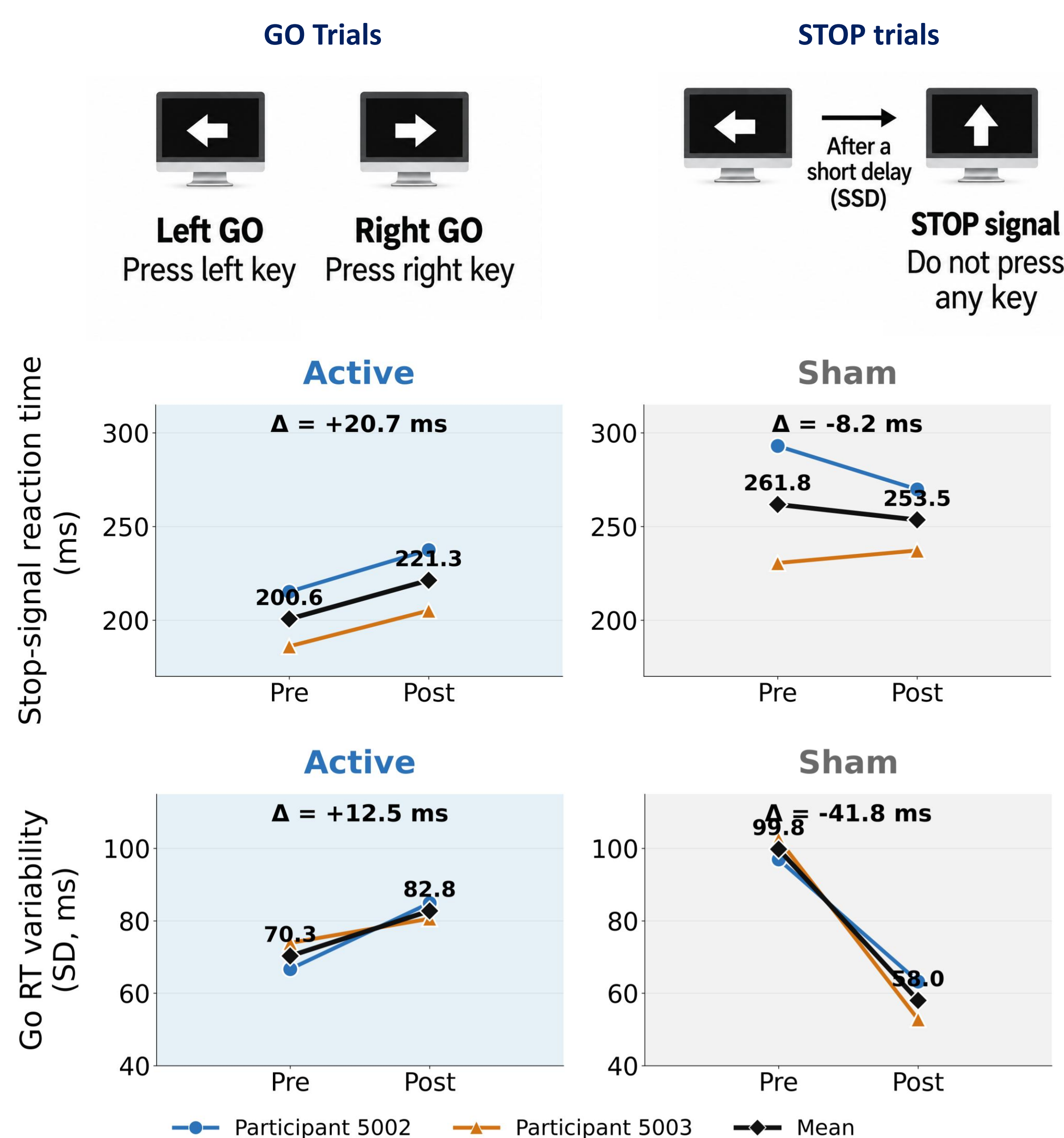


Figure 1. SST performance before and after active and sham LIFU. Colored lines show individual participants and black lines show means. Active LIFU was followed by a consistent increase in stop-signal reaction time (SSRT; mean $\Delta = +20.7$ ms), whereas sham showed no consistent increase (mean $\Delta = -8.3$ ms). GO trial reaction time variability increased slightly following active LIFU (mean Δ SD = +12.5 ms), while both participants showed reduced variability following sham (mean Δ SD = -41.8 ms) (n=2). Higher SSRT indicates reduced response inhibition. Higher Go RT variability indicates reduced response consistency.

Behavioral: Beat-Based Advantage Task (BBA: Timing)

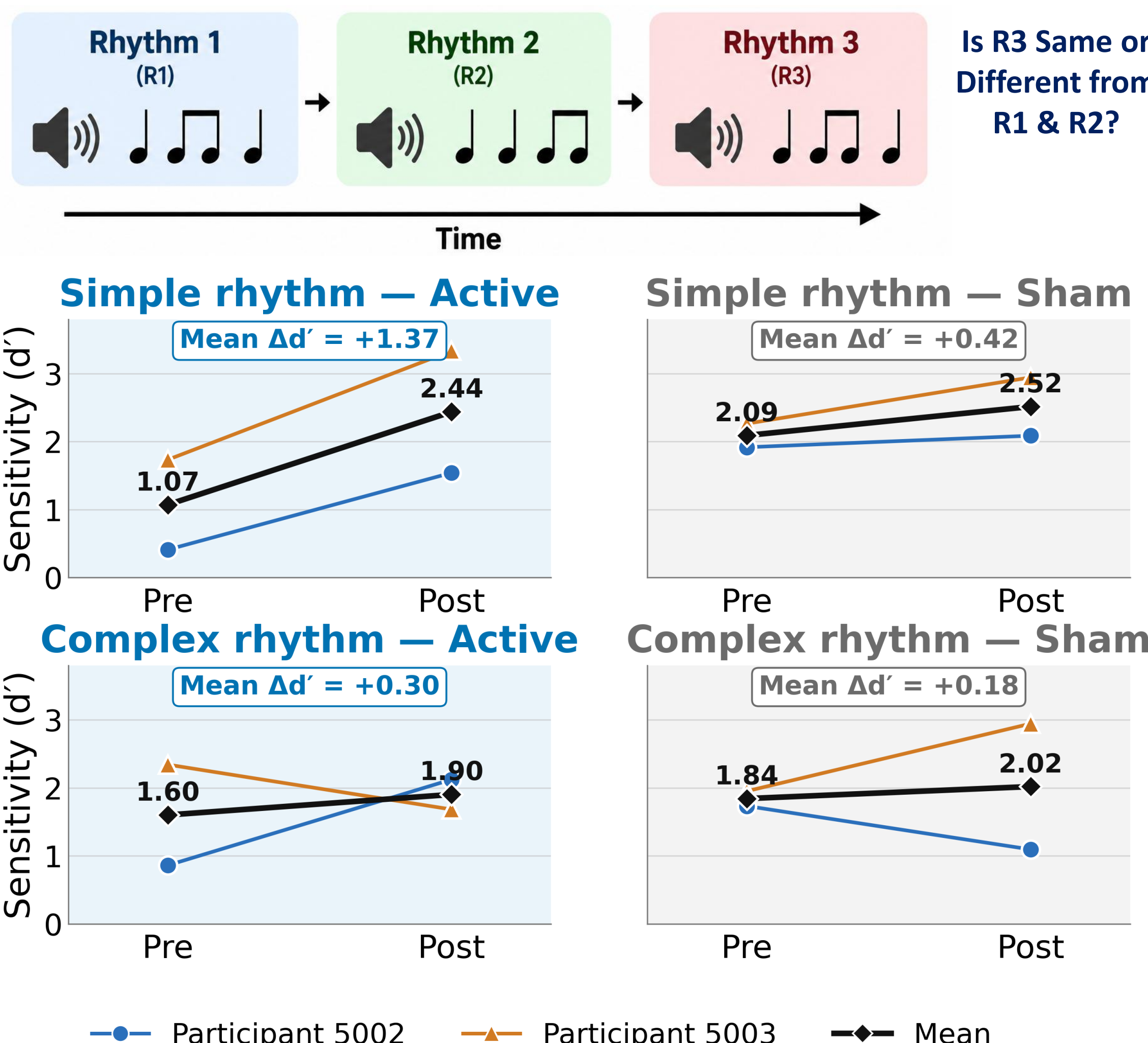


Figure 2. Rhythm sensitivity before and after Active and Sham LIFU. Colored lines show individual participants and black lines show means. Simple-rhythm discrimination improved more after Active compared to Sham (mean $\Delta d' = +1.37$ vs +0.42), whereas complex-rhythm changes were heterogeneous (mean $\Delta d' = +0.30$ vs +0.18). Higher d' indicates better discrimination (n = 2).

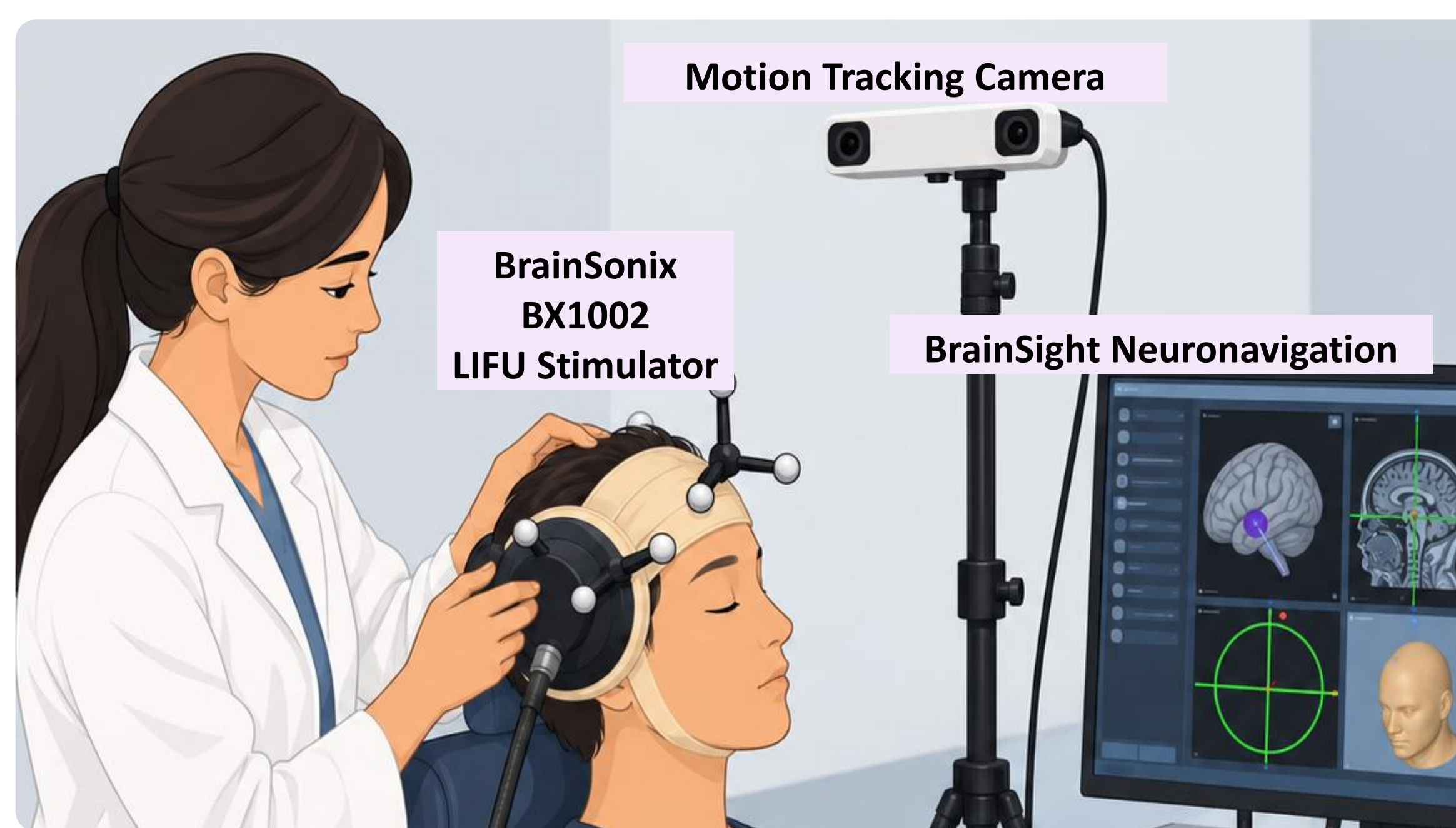


Figure 3: Schematic illustration of the stimulation experimental setup, generated with assistance from OpenAI.

ClinicalTrials.gov



OSF Preregistration

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NCT07668882



LIFU Stimulation: BrainSonix transducer/BXPulser 1002; 650 kHz, 10 Hz PRF, 100 ms pulses, 70% duty cycle, 30 s ON / 30 s OFF, targeted to the left VIM.



Neural: resting-state fMRI (Functional Connectivity)

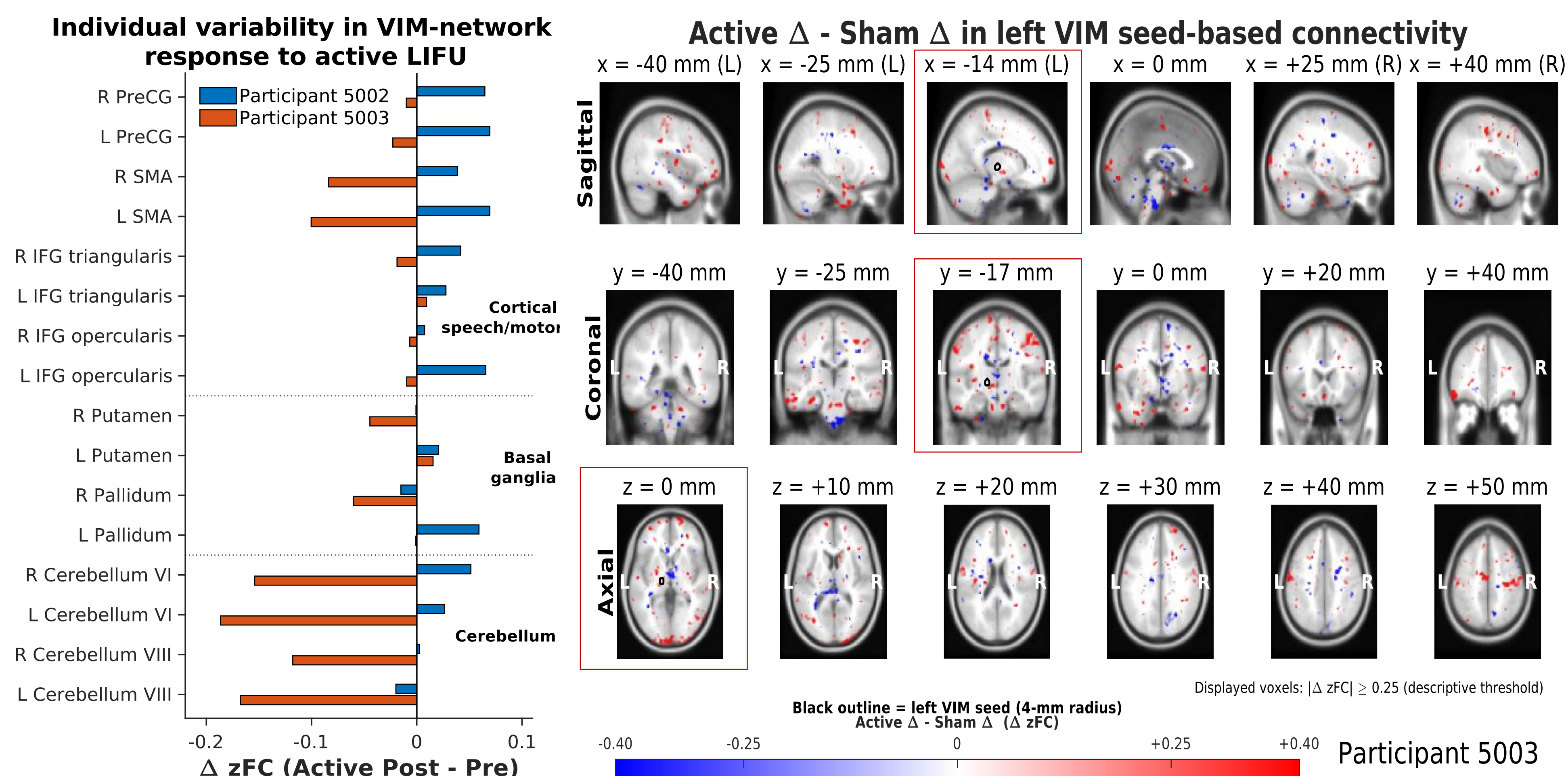
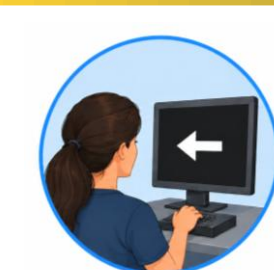


Figure 4. Individual active-LIFU responses in VIM-network connectivity. Pre-to-post changes in left VIM seed-based resting-state functional connectivity (Δ zFC = Active Post - Pre) across speech/motor cortical, basal ganglia, and cerebellar ROIs in participants 5002 and 5003. Values are descriptive and illustrate individual variability in connectivity changes following active stimulation.

Figure 5. Active-versus-sham modulation of VIM-centered connectivity. Whole-brain left VIM seed-based connectivity contrast for participant 5003 only (Active Δ - Sham Δ ; Δ = Post - Pre). Red indicates a more positive pre-to-post change following active relative to sham stimulation, whereas blue indicates a more negative change. The black outline denotes the 4-mm left VIM seed. Results are displayed descriptively at $|\Delta$ zFC| ≥ 0.25 ; no inferential statistical threshold was applied.

CONCLUSION



SST Behavioral → Preliminary findings suggest that active LIFU may **modulate response-control dynamics, attenuating the response stabilization** observed following sham stimulation (In the opposing direction of H3 hypothesis).



BBA Behavioral → Preliminary findings suggest that active LIFU may **selectively enhance simple-rhythm discrimination** relative to sham (as hypothesized in H4), while complex-rhythm responses remain heterogeneous.



rsfMRI neural → Preliminary findings remain heterogeneous across participants.

Recruitment and data collection are ongoing. Here, we present preliminary results from n = 2 participants. The study aims to establish an individualized workflow that integrates acoustic modeling, MRI-guided targeting, LIFU neuromodulation, and multimodal (neuroimaging, behavioral, and speech) assessment to probe the causal thalamocortical mechanisms underlying PDS.

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